Summary: Leiter, Neumann, Egusa, Harashina & Hasenauer (2022), "Assessing the Resource Potential of Mountainous Forests: A Comparison between Austria and Japan"

Source: *Forests* 2022, 13, 891. https://doi.org/10.3390/f13060891 (open access, CC BY). Uses Japanese National Forest Inventory (JNFI, 2004-2013) and Austrian National Forest Inventory (ANFI, 2000-2018) plot data.

Headline finding

Japan's low wood self-sufficiency (~35% in 2016) is **not** explained by lower forest growth rates — annual increment per age class in Japan is statistically similar to Austria's (Welch t-test, most age classes NS). The real gap is **utilization**: Austria harvests 88% of annual increment vs. Japan only 53-58%. Japan is sitting on a large, growing, under-harvested timber stock.

Key comparative facts

	Japan	Austria
Forest coverage	~66% of land (25M ha)	~50% of land (4.4M ha)
Forest area trend since 1970	~flat (slight decline)	+4%
Total standing volume	3.14bn m³ (1990) → 4.9bn m³ (2012)	0.99bn m³ (1996) → 1.17bn m³ (2016)
Median stocking volume/ha	393 m³/ha (plantation)	281 m³/ha
Harvest-to-increment (utilization) rate	32% (2001) → 58% (2011)	steady rise to 88% (2018)
Forest road density	14 m/ha	45 m/ha
Dominant species	Sugi (*Cryptomeria japonica*, 35.7% vol.), Hinoki (15.2%)	Norway spruce (*Picea abies*, 60.4% vol.)
Dominant age class	41-60 yr (~50% of standing volume — badly skewed)	41-60 yr but only ~20% of volume (much more balanced)
2018 wood demand vs. domestic production	Demand 82.5M m³; domestic 30.2M m³ (37%); imports 52.3M m³	Demand met mostly domestically; 25.5M m³ produced, 12.8M m³ imported (largely re-exported as sawn wood)

Why the increment finding matters

Prior Japanese studies (e.g., Kuboyama et al.) assumed Austria's higher stocking-per-hectare and infrastructure explained the productivity gap. This paper's more granular NFI-plot analysis overturns that: growth potential is comparable or even slightly higher in Japan per hectare. The bottleneck is structural/economic — forest road density, aging/absentee forest owners, fragmented small-scale ownership, and low profitability — not biology.

The age-class imbalance problem

Japan's post-war afforestation boom (peaking 1953 and 1961) created a huge cohort now aged 40-60+, all becoming harvestable at once with little regeneration behind it (Fig. 7). Austria's "Ertragswald" (managed forest, 80% of Austrian forest area) has a much flatter age distribution — the result of consistently higher utilization sustained over decades. Without action, Japan faces either (a) a harvesting bottleneck as the entire post-war cohort becomes over-mature simultaneously, or (b) continued underuse and stand collapse/economic loss.

Species regeneration asymmetry

Austria's main species (Norway spruce) regenerates naturally; Japan's main species (sugi) does not regenerate well naturally, and its second species (hinoki) barely regenerates under canopy. This creates a structural split between "planted forest" and "natural forest" in Japan that doesn't exist in Austria, reinforcing reliance on active replanting after harvest.

Authors' conclusion

Japan has high near-term harvesting potential from its existing over-aged stock — increasing harvest rate could raise self-sufficiency and create rural income — but the priority for the next decade is achieving a **balanced age structure** via continuous replanting after harvest, alongside investment in forest roads and management knowledge transfer to (aging, often disengaged) forest owners.

Relevance to Mitsue project

Strong empirical backing for the CHP fuel-supply thesis in [[reference_fit_grid_biomass]] and [[project_sugano_fuel_partner]]: Japan nationally (and by extension Mitsue/Sugano's Sugi-dominant, aging plantations) is *not* growth-constrained — it is harvest- and infrastructure-constrained. This directly supports the "worker shortage not land/wood supply is the bottleneck" framing in [[project_council_minutes_forestry_findings]] and strengthens any subsidy/pitch argument that a CHP+DC anchor customer for low-value harvested wood would help correct a genuinely overstocked, mismanaged national forest resource rather than compete for a scarce one. Worth citing directly (with the 53-58% utilization figure) in outreach decks.